

SCI-JINC

Engineering Conformance Specification

Contract v0.1 / document r0.3

2026-08-29

Scientific owner: Grant Wilson

Status. Frozen conditional implementation-independent scientific authority, SCI-JINC v0.1/r0.3. This document expresses the same shared authority as the companion rationale and binds SCI-JINC-DEC-PHASE-CENTER-001. It contains no implementation mapping and assesses no implementation candidate. No TolTEC numerical JINC route is authorized until every separately owned input named by this contract is available. Compilation, source-byte checks, and specification replay are not evidence of implementation conformity, representation or response/covariance fidelity, parameter adequacy, validation, achieved performance, readiness, production, or production authorization. Any scientific correction requires a versioned successor; these frozen bytes are immutable.

Contract identity	SCI-JINC v0.1
Document identity	Engineering Conformance Specification r0.3
Companion view	Scientific Rationale and Contract v0.1/r0.3
Normative core	Six shared modules, included once in identical order in both views
Requirements	SCI-JINC-REQ-001 through SCI-JINC-REQ-044, sequential with no gaps
Predictions	SCI-JINC-PRED-001 through SCI-JINC-PRED-036, sequential with no gaps
Present assessment	Not assessed ; no implementation candidate, configuration, product, test, audit, validation result, or reduction was inspected

Contents

1 Purpose and result vocabulary	3
2 Scientific-source byte closure	3
3 ODQ-107 scope-parity control	4
4 Applicability and fail-closed inputs	4
5 Exact discrete reference oracle	5
6 Numerical profile and certificate	6
7 Compact generative replay	6
8 Future assessment and evidence	7
9 Output-role and claim discipline	7
10 Canonical notation and units	8
11 Normative definitions and boundary	9
12 Canonical equations	11

12.1 Peak-normalized analytic family	11
12.2 Exact discrete point-phase and WCS operator	11
12.3 Signed estimator	12
12.4 Conditioning and local support	13
12.5 Coefficient-squared temporal accounting	13
12.6 Conditional mathematics retained for future science	13
13 Assumptions, validity classes, and explicit unavailability	14
14 Binding scientific requirements	15
15 Falsifiable predictions and edge cases	19
15.1 Prediction-to-requirement trace	22
16 Blank conformance-result template	23

1 Purpose and result vocabulary

This specification turns the shared SCI-JINC authority into a future conformance-review surface and introduces no independent science. If this engineering prose is broader than the shared modules, the shared module controls. A future result is exactly *pass*, *fail*, *not assessed*, or *not applicable under an explicit contract-authorized condition*. Missing evidence is never *pass*.

The conformance object is one claimed realization under one exact contract, source packet, observation, array, plan, coefficient family, parameter set, target WCS, destination, numerical profile, certificate, and lifecycle generation. These identities must be bound before any conclusion.

2 Scientific-source byte closure

The Stage B author verified the manifest digest and all sixteen admitted object digests before authorship. A later assessor must repeat the checks from independently obtained bytes. Byte identity alone establishes neither candidate conformity nor scientific validity.

Table 1: Required exact source checks.

Object	Expected binding	Freeze check
Author-packet manifest	Commit 88dcce8b0f7b1d78053b25831b39cf370afd47cc; SHA-256 52a8e843456a8cb033b7593d9b9f67fb83b0ee565c91c141d8e16d46b906140e.	Byte-equal
Targeted r0.2 repair directive	SHA-256 c07505861d91459f69e7d0989f11551e2a14265c916cd5772ea48a86bb186ed2.	Byte-equal
Final targeted r0.3 directive	SHA-256 4878e1745e085b4e33d2e71f1190299d72f2cfd7b2215e36a9e8405a977bd207.	Byte-equal
Final owner-review/freeze directive	SHA-256 958cffeac67c11e916527c0f78e9c80d648f68d5eec38a0607fc4af1511dddec.	Byte-equal
Owner phase/center decision	SCI-JINC-DEC-PHASE-CENTER-001. Phase statement “I approve disposition A”, SHA-256 9a70cbc63c0c79a7db70ad9796481fb0fe3f1f4c2d7524820b54cec68b8b1620. Separate center statement exactly as submitted, SHA-256 3b79351f7661e2432a5426fba3a16e9710c2fae0b34fd9b8f60dd45bca837ecb.	Bound separately; no inference
Manifest objects	Exactly sixteen entries, each equal to its manifest SHA-256 before use.	16/16
Successor Scope Brief	SHA-256 5f2505c2760fc5cb07506249f33f449651aca67cccc9c444305b059674f0ddbd.	Byte-equal
PTC boundary	SCI-PTC_TO_SCI-JINC v0.1/r0.3; SHA-256 5769d413460e931745e0d401ea432b12d1077c15466247c49caa71b997d4ab1e.	Byte-equal
AST boundary	SCI-AST_TO_SCI-JINC v0.1/r0.2; SHA-256 efffa7059b59c89793fa1d523fb3bb48235f1ab55f7d55060af1600cbfd470a5.	Byte-equal
JINC upstream-occurrence admission	Immutable Registry identity SCI-JINC:jinc_map_contribution@1; SHA-256 2db95da7e5d1b980df79993907d45ac0ababc3aa05c189bfb62dcf04ff2c2e8a.	Byte-equal

Object	Expected binding	Freeze check
VAL binding and successors	Binding SHA-256 ee8f20db5febdb51e39f7157449d6c2d 03a0d17058605dd5531e9ab5ca439e30; SCI-VAL_SOURCE_BINDING_REGISTER successor v0.1/ r0.3-map-r0.7.1-jinc-stage-a-q002-2026-08-28, SHA-256 0e7ca29ee2e9cd02fb1b76cf87cc64fc e6164407a7801f9b9a105ca646317e88; SCI-VAL_PROFILE_REGISTRY successor with the same version, SHA-256 4b9a1ebecfc847c83b59da772afd9b03 1ab1830e8febbb12d1a47f70ce5a1110.	Exact process bindings under the base-v0.1 exact-source lock

No unlisted package file, implementation, schema, configuration, test, audit, validation product, reduction, status document, or web source may fill a gap. The complete exact PTC, AST, profile, binding, and JINC-specific SCI-VAL successor snapshot files named above are locked sources. No ambient current Registry substitution is allowed; a changed bound object requires a versioned SCI-JINC successor.

3 ODQ-107 scope-parity control

The admitted successor Scope Brief records an owner-approved ODQ-107 amendment that closes base v0.1 to five numerical roles and defers response, covariance, formal weight, generalized companion availability, and generalized provenance products. The pre-repair scope therefore matched its admitted Stage A authority. The targeted directive's conditional broadening branch does not apply because that exact owner amendment already exists. No new amendment is needed to retain REQ-038-040. Any future broadening needs a bounded successor Stage A amendment and renewed exact-byte owner approval. No candidate may add those roles or interpret their absence as zero, independence, or achieved fidelity.

Table 2: Deferred response and covariance distinctions.

Family	Preserved science	Base-v0.1 state
Fixed-state response	Apply the fixed normalized operator once to one exact available processed response companion.	Conditional mathematics only
PTC procedure response	Re-run PTC while holding the JINC operator fixed.	Deferred
JINC re-resolved response	Re-resolve admission, coefficient, parameter, phase, support, cancellation, or destination state.	Deferred and distinct
Whole-chain response	Complete RTC-CAL-PTC-JINC rerun under separate authority.	Unavailable absent authority and rerun
Covariance	$\mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^T$ on exact available domains.	Conditional; missing is not zero or independence
Diagonal variance/weight	Q_p/C_p^2 and C_p^2/Q_p under an exact true-inverse-variance family and declared independence.	Conditional; no base product

4 Applicability and fail-closed inputs

A numerical route requires mutually compatible exact bindings for:

1. the transformed occurrence and same-processed-sample AST coordinate;

2. one registered, selected, realized PTC family that explicitly permits SCI-JINC, with payload, QC, generation, and source identity;
3. one owner-authorized TolTEC array parameter set;
4. one positive-integer phase count selected by the accepted request or one exact owner-authorized versioned mode policy;
5. one compatible target WCS and positive sample frequency; and
6. one exact numerical profile and matching certificate when numerical adequacy or near-cancellation support is claimed.

The coefficient family/payload, TolTEC parameter set, numerical profile, and certificate are not admitted or supplied by this source packet. Their routes are typed unavailable. Unity, ordinary-MAP permission, another family, an inherited 45 m denominator, inherited shape values, mode-dependent $r_{\max}$, an ambient phase count, or any implementation default cannot repair them.

5 Exact discrete reference oracle

The oracle evaluates the midpoint-quantized discrete operator in exact real arithmetic. There is no hidden continuous-phase or pixel-area-integrated target.

For $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$ and exact AST coordinate $\mathbf{u}_i$, resolve componentwise

$$c_{i\mu} = \lfloor u_{i\mu} + \tfrac{1}{2} \rfloor, \quad \phi_{i\mu} = u_{i\mu} - c_{i\mu} \in [-\tfrac{1}{2}, \tfrac{1}{2}).$$

Decision **SCI-JINC-DEC-PHASE-CENTER-001** selects the larger integer at the positive-axis half tie. For $n_{\text{sub}} = \text{subpixel_n} \in \mathbb{N}_{>0}$,

$$q_{i\mu} = \lfloor n_{\text{sub}}(\phi_{i\mu} + \tfrac{1}{2}) \rfloor, \quad \hat{\phi}_{i\mu} = -\tfrac{1}{2} + \frac{q_{i\mu} + \tfrac{1}{2}}{n_{\text{sub}}}.$$

Bins are left closed and right open, and an interior-boundary tie selects the upper bin. No parity restriction exists. With even n_{sub} , exact $\phi = 0$ maps to $+1/(2n_{\text{sub}})$. The same decision retains this rule for every positive n_{sub} with no zero-phase special case. WCS axis order and orientation are normative. The logical cache key is $(q_{i1}, q_{i2}, d_1, d_2)$; storage may differ only by a proven value-preserving bijection.

The complete target WCS binds frame, projection, reference values, one-based reference pixel, dimensions, units, axis order/signs, rotation, handedness, version, and constant pixel matrix $\mathbf{M}$, with

$$\mathbf{M}^T \mathbf{M} = \Delta^2 \mathbf{I}, \quad \Delta > 0.$$

Arbitrary recorded rotation or reflection is permitted. Skewed or anisotropic geometry is incompatible; a scalar size cannot conceal it.

For array a ,

$$h_a = \lceil s_a(r_{\max})_a / \Delta \rceil, \quad \mathbf{d} \in \{-h_a, \dots, +h_a\}^2.$$

An occurrence with $\mathbf{c}_i \notin \mathcal{D}$ contributes nowhere. Otherwise set $\mathbf{p} = \mathbf{c}_i + \mathbf{d}$ and retain it exactly when $\mathbf{p} \in \mathcal{D}$. There is no circular cutoff, wrap, reflection, completion, or edge renormalization. At the destination pixel centre,

$$\hat{r}_{i\mathbf{d}} = \|\mathbf{M}(\mathbf{d} - \hat{\phi}_i)\|_2, \quad \hat{r}'_{i\mathbf{d}} = \hat{r}_{i\mathbf{d}} / s_a.$$

The exact second-factor root is $j_{1,1} = \min\{x > 0 : J_1(x) = 0\}$; a rounded decimal is nonnormative. The shared kernel and estimator equations then control. Outer rows and corners remain evaluated beyond the first-zero radius.

For an exact integer-center occurrence with even n_{sub} , zero phase on both axes and $\mathbf{d} = \mathbf{0}$ shall give

$$\hat{r}_{i\mathbf{0}} = \frac{\Delta}{\sqrt{2}n_{\text{sub}}}, \quad \kappa_{i,\mathbf{c}_i} = \kappa_a \left(\frac{\Delta}{\sqrt{2}n_{\text{sub}}s_a}; a_a, b_a, c_a, (r_{\max})_a \right).$$

The expected coefficient is the exact nonzero-radius oracle value, not an imposed $\kappa_a(0) = 1$.

6 Numerical profile and certificate

Algebraic support requires finite inputs and exact discrete coefficients, $Q_p > 0$, and $C_p \neq 0$. Numerical adequacy separately requires the complete pair below.

Table 3: Profile-plus-certificate checklist.

Field	Required exact content
Profile identity	Owner, identifier, version, source digest, compatibility, lifecycle, and supersession.
Observable and error	Observable; absolute, relative, or mixed form; reference norm/magnitude; units; numerator and denominator treatment.
Domain and aggregation	Exact source/parameter/pixel domain and simultaneous, componentwise, or worst-case interpretation.
Near-zero and inequality	Exact zero/near-zero rule and one exact pass/fail inequality, without a qualitative fidelity substitute.
Certificate method	Exact oracle, bound/comparison method, uncertainty semantics, and authorized coverage.
Certificate binding	Candidate, source bytes, observation/array, coefficient generation, parameters, WCS, phase count, destination, reduction policy, profile, lifecycle, and immutable certificate identity.
Certificate result	Exact bounds, exceptions, and pass/fail over the complete declared domain.

If $|\hat{C}_p - C_p| \leq \epsilon_{C,p}$ is certified, normalized support requires $|\hat{C}_p| > \epsilon_{C,p}$. Otherwise zero remains inside the certified denominator interval. The profile must provide any additional map-error inequality. A profile alone, certificate alone, successful run, two agreeing realizations, or finite nonzero machine result is not a pass.

A passing pair establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle on its declared domain. It does not establish parameter or phase-count adequacy, square-truncation adequacy, source-response or beam fidelity, photometric or astrometric performance, optimality, observational validation, response/covariance fidelity, readiness, or production authority. TolTEC parameter-set qualification is a separate scientific and observational program.

No exact owner-approved profile or matching certificate is admitted or supplied. Numerical adequacy and near-cancellation support are therefore **typed unavailable**, not tested, failed, or achieved.

7 Compact generative replay

A future assessor shall:

1. verify all source bytes and bind requested, effective, resolved, and realized contract, plan, observation, array, destination, and lifecycle;
2. reconstruct the occurrence population from immutable PTC/AST parents and JINC upstream-occurrence admission under its exact Registry identity;
3. bind coefficient permission/generation/payload/QC, parameter set, WCS metric, phase count, and sample frequency;
4. reapply the owner-approved upper-bin lattice and positive-axis tie, bins, midpoints, square extent, centre gate, edge crop, and point-centre radius;
5. evaluate the exact discrete kernel and coupled N_p , C_p , Q_p , m_p , and $T_p^{(\kappa^2)}$ equations;
6. evaluate the exact profile/certificate and denominator separation;
7. verify five-role atomic publication, local versus whole failure, the static prohibition on companion roles, completion, and lifecycle; and
8. compare regenerated identities and values under the profile's exact relation, recording every mismatch without fallback.

The compact generative record binds every item above, including source, parentage, coefficient, parameter, WCS, geometry, cancellation, profile/certificate, destination, ODQ-107 scope identity, exact-source lock, required-role state, failure, and completion. It contains no per-bundle response/covariance information-state record. It is plan/bundle lineage, not a generalized provenance product or dense per-contribution archive.

8 Future assessment and evidence

A later assessor binds identity and applicability, executes the replay, checks units, signs, limits, grouping, lifecycle, and five-role publication, then evaluates all 44 requirements and 36 predictions. Each record contains construction, expected and observed results, evidence locator, exact profile/tolerance authority, and result state. Numerical assessment occurs only when both profile and certificate are authorized and bound.

Table 4: Minimum evidence record.

Category	Minimum exact content
Authority and boundaries	Contract/source digests, candidate/lifecycle, PTC coefficient and same-sample AST coordinate parentage.
Discrete operator	FITS coordinate, tie/bin/midpoint rules, WCS metric, Bessel root, square extent, point centre, and edge membership.
Algebra and adequacy	Coupled accumulators, units, limits, exact profile, certificate, oracle, domain, error functional, and denominator separation.
Bundle and replay	Generative record, array cardinality, destination, chunking, five-role atomicity, and no cross-observation route.
Deferred companions	No unauthorized response, covariance, formal-weight, exposure, availability, diagnostic, or provenance role; missing covariance is not zero.
Results and claims	All stable identifiers plus separate states for conformity, representation, response/covariance fidelity, validation, performance, readiness, and production.

This is evidence ordering, not implementation architecture. Compilation, a source hash, one numerical example, or one stored product cannot establish complete conformity.

9 Output-role and claim discipline

The ODQ-107 schema remains exactly N_p , C_p , Q_p , derived m_p with local support, and $T_p^{(\kappa^2)}$. Q_p is not automatically precision; C_p is not exposure; coefficient-squared time is not acquired time. Whole role failure suppresses the bundle, while local cancellation or unavailable support remains map content without a finite substitute.

Algebraic conformity, representation fidelity, response/covariance fidelity, numerical validation, observational performance, readiness, and production authorization are separate conclusions with separate evidence and authority. At revision r0.3, no candidate was inspected and every later claim is not assessed, not performed, or no claim.

10 Canonical notation and units

The symbols in Table 5 are normative in both views of this contract. The generic signal unit is denoted by U . The admitted upstream boundary presently names a nonpolarimetric total-intensity-equivalent quantity in mJy/beam ; the symbol U keeps the algebra explicit and does not turn a component token into a physical Stokes assertion.

Table 5: Canonical notation.

Symbol	Meaning and domain	Unit
a	Stable TolTEC array identity, exactly one of a1100 , a1400 , or a2000 .	identity
i	Exact admitted PTC occurrence on stable RTC output sample n , with observation, detector occurrence/UID, PTC application generation, segment, array, and immutable parents.	identity
p	Target pixel in one exact JINC destination WCS.	identity
$\mathbf{u}_i = (x_i, y_i)$	Exact finite AST continuous FITS pixel coordinate for occurrence i in the target JINC WCS.	pixel
$\mathbf{c}_i$	Integer FITS pixel-center coordinate selected from $\mathbf{u}_i$ by Equation (4).	pixel
ϕ_i	Exact residual phase $\mathbf{u}_i - \mathbf{c}_i$ in $[-\frac{1}{2}, \frac{1}{2})^2$.	pixel
n_{sub}	Effective integer phase resolution represented by subpixel_n ; every $n_{\text{sub}} \geq 1$ is permitted under the owner-approved upper-bin phase-lattice rule in SCI-JINC-DEC-PHASE-CENTER-001 .	1
$\mathbf{q}_i$	Two-axis phase-bin index in $\{0, \dots, n_{\text{sub}} - 1\}^2$.	index
ϕ_i	Midpoint representative of the exact half-open phase bin selected by $\mathbf{q}_i$; for even n_{sub} , exact zero phase selects $+1/(2n_{\text{sub}})$ on each applicable WCS axis.	pixel
β_i	Exact axis-1-fast linear phase index $q_{i1} + n_{\text{sub}}q_{i2}$.	index
$\mathcal{K}_{id}$	Logical coefficient-cache key (β_i, d_1, d_2) , equivalent to $(q_{i1}, q_{i2}, d_1, d_2)$.	identity
$\mathbf{M}$	Exact two-by-two affine tangent-plane pixel matrix of the target WCS, including axis order, signs, rotation, and handedness.	angle per pixel
Δ	Positive target-pixel angular scale satisfying $\mathbf{M}^T \mathbf{M} = \Delta^2 \mathbf{I}$ for the base-v0.1 square-pixel WCS class.	angle per pixel
$\mathcal{D}$	Finite destination pixel-center domain $\{1, \dots, N_x\} \times \{1, \dots, N_y\}$.	identity
h_a	Integer square-cache half-width $\lceil s_a(r_{\text{max}})_a / \Delta \rceil$.	pixel
$\mathbf{d}$	Integer cache offset in $\{-h_a, \dots, h_a\}^2$.	pixel
z_i	Exact finite transformed signal Z_i^{PTC} .	U
ω_i	Positive, finite, PTC-produced analysis/gridding coefficient from one exact registered family/version that permits SCI-JINC.	1 or U^{-2} , as declared by its family
κ_{ip}	Signed, dimensionless analytic JINC coefficient at the selected point phase. Finite positive, exact-zero, and finite negative values are distinct legitimate analytic results.	1
w_{ip}	Complete signed contribution coefficient, $w_{ip} = \omega_i \kappa_{ip}$.	unit of ω_i
I_{ip}	Boolean membership after sample admission, exact AST association, rounded-center domain admission, local square geometry, and all pixel-local gates.	1
s_a	Explicit angular radial scale associated with stable array a ; finite and strictly positive in any numerical realization.	angle

Symbol	Meaning and domain	Unit
r	Generic angular separation used by the analytic family.	angle
$\hat{r}_{id}$	Exact discrete angular separation between the representative sample phase and destination pixel center for cache offset $\mathbf{d}$, from Equation (13).	angle
r'_a	Dimensionless radius r/s_a , finite and nonnegative.	1
$j_{1,1}$	Exact first positive zero $\min\{x > 0 : J_1(x) = 0\}$ of the first-order Bessel function.	1
$a_a, b_a, c_a, (r_{\max})_a$	Ordered, dimensionless analytic shape tuple for array a ; every member is finite and strictly positive in an authorized numerical realization.	1
θ_a	Complete kernel parameter identity $(s_a, a_a, b_a, c_a, (r_{\max})_a)$, including owner, source, version, units, array association, and lifecycle state.	mixed, as above
N_p	Signed signal numerator.	$U[\omega]$
C_p	Signed normalization; it may be finite and negative, but exact zero is never accepted.	$[\omega]$
Q_p	Quadratic accumulator; finite and strictly positive on accepted local support.	$[\omega]$
m_p	Normalized estimate N_p/C_p on accepted local support.	U
D_p	Absolute-term sum $\sum_i I_{ip} \omega_i\kappa_{ip} $, used only as construction state for conditioning.	$[\omega]$
ρ_p	Dimensionless cancellation statistic $ C_p /D_p$.	1
$\mathcal{P}_{NA}$	Exact owner-approved numerical-adequacy profile, including its observable, norm, domain, error treatment, exact inequality, uncertainty rule, lifecycle, provenance, and explicit exact-discrete-oracle-only claim boundary.	identity
$\mathcal{E}_p$	Exact numerical-adequacy certificate for pixel p under $\mathcal{P}_{NA}$; a passing certificate establishes only the declared finite-precision fidelity to the exact selected discrete oracle.	identity
G_a	Compact bundle-level generative record for one observation and array; it identifies immutable parents, selections, joins, operator state, lifecycle, and completion without storing dense per-contribution provenance.	identity
A_{pi}	Fixed-state normalized linear operator row $I_{ip}\omega_i\kappa_{ip}/C_p$.	1
$f_{s,i}$	Effective processed-sample frequency, finite and strictly positive.	s^{-1}
$T_p^{(\kappa^2)}$	Required <code>jinc_coefficient_squared_time</code> coefficient-squared temporal-accounting accumulator.	s

In-memory indices are zero-based. Persisted FITS/WCS pixel coordinates and the mathematical coordinates in Equations (4)– (13) are one-based at integer pixel centers. Memory order never supplies axis sign, handedness, orientation, or wrapping.

11 Normative definitions and boundary

SCI-JINC observation estimator.

A signed spatial-kernel transformer that consumes exact PTC signal occurrences and their exact AST coordinates and forms at most one complete JINC bundle for each admitted and requested stable array in one observation. It is a sibling alternative to ordinary SCI-MAP, not a downstream stage of SCI-MAP.

Exact occurrence.

The scientific identity that jointly fixes observation, detector occurrence and UID, stable RTC output sample, PTC

product and application generation, PTC segment, stable array/group, time attribute, and immutable parent chain. Row position, nearest time, tolerance, shape, detector order, or numerical equality is not a substitute.

Exact AST association.

The relation binding z_i to the AST-owned `SCI-AST:rtc_output_grid_coordinates@1` continuous coordinate for the same processed sample realization in the exact target JINC WCS. Its data-model join is an engineering choice; its scientific identity is not.

JINC upstream-occurrence admission.

The canonical human-readable term for the decision that one exact upstream occurrence may be considered by JINC. Its immutable Registry identity is `SCI-JINC:jinc_map_contribution@1`; the identity remains fixed, but its name shall not be used as shorthand for arithmetic contribution. This decision neither authorizes rounded-center admission, sample-pixel membership, cancellation support, numerical contribution, nor bundle publication.

Rounded-center domain admission.

The JINC-owned occurrence-level gate applied after center rounding and before any footprint evaluation. The resolved rounded center must lie within the finite destination pixel domain. An outside center makes $I_{ip} = 0$ for every p , even when the square footprint would overlap the map.

Point phase.

The residual between the exact continuous sample coordinate and the exact rounded center of Equation (4). Equations (6)–(8) select a half-open bin and its midpoint representative independently on each axis. The resulting kernel matrix is point evaluated and is not a pixel-area average. Under decision `SCI-JINC-DEC-PHASE-CENTER-001`, every positive n_{sub} remains permitted. For even n_{sub} , exact zero phase selects the positive midpoint on each applicable WCS axis; no zero-phase special case replaces that representative. The positive-axis half-pixel center tie is likewise intentional.

Square support.

The fully populated integer-offset cache of Equation (12), with exact half-width $h_a = \lceil s_a(r_{\text{max}})_a / \Delta \rceil$. Every resolved square pixel is evaluated, including corners with radius larger than $(r_{\text{max}})_a$; there is no circular membership predicate.

Sample-pixel membership.

The Boolean proposition $I_{ip} = 1$ after sample admission, exact coordinate association, rounded-center domain admission, square placement, finite local geometry, and every pixel-local precondition. A contract-defined analytic zero is a member with zero contribution; outside support is nonmembership; a finite negative coefficient is an ordinary signed member.

Algebraic local support.

The exact-arithmetic state in which the required inputs and accumulators are finite, $Q_p > 0$, and $C_p \neq 0$. It establishes that N_p/C_p is algebraically defined; it does not certify a finite-precision realization.

Certified numerical-realization adequacy.

The state in which an exact owner-approved profile $\mathcal{P}_{\text{NA}}$ and a matching certificate $\mathcal{E}_p$ bind the scientific observable, absolute or relative form, reference norm, parameter/source/pixel domain, numerator and denominator error treatment, near-zero behavior, simultaneous or worst-case interpretation, exact pass/fail inequality, uncertainty, lifecycle, and provenance, and the certificate passes that inequality. No such profile or certificate is admitted or supplied by this source packet, so numerical-realization adequacy is presently typed unavailable rather than approximated by prose. A passing pair establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over its declared domain. It does not qualify the selected parameters, phase resolution, square truncation, scientific response, performance, validation, readiness, or production use.

Exact-source lock.

The base-v0.1 compatibility rule that binds the complete exact PTC boundary, AST boundary, admission-profile object, SCI-VAL binding, and JINC-specific source/profile successor snapshot bytes named by `SCI-JINC-REQ-043`. Ambient or merely current Registry state is not a substitute. Any changed bound object requires a versioned SCI-JINC successor; numerical or textual similarity cannot establish compatibility.

Complete observation bundle.

The atomic fixed-role product for one observation, stable array, exact JINC plan and realization, target WCS, and lifecycle generation. It contains exactly five numerical map-plane roles: N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$. Its compact bundle-level generative record G_a is required information state at the existing plan/bundle provenance granularity; it is not a sixth product role, a dense per-contribution archive, or generalized provenance machinery.

Whole-bundle failure.

Failure to form or publish any required whole-product role. It suppresses the affected bundle without a partial

role, placeholder, synthetic empty-array product, or completion override. Pixel-local invalid support is ordinary bundle content and is not whole-bundle failure.

Typed numerical unavailability.

The fail-closed state in which a numerical JINC route cannot be realized because an exact authorized TolTEC kernel parameter set or an exact registered, selected, realized, JINC-permitted PTC coefficient family/payload is absent. The same state applies when an exact owner-approved numerical-adequacy profile and matching certificate are required but absent. It is not repaired by a hidden default, unity coefficient, alternate family, ordinary MAP permission, inherited scale, implementation value, or approximate fidelity phrase.

Scientific destination identity.

The product identity formed from observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Worker, thread, process, container, filename, vector position, and network order are not scientific identity.

12 Canonical equations

12.1 Peak-normalized analytic family

Let J_1 be the first-order Bessel function of the first kind, and define its first positive zero exactly by

$$j_{1,1} = \min\{x > 0 : J_1(x) = 0\}. \quad (1)$$

The rounded decimal 3.831706 is only a nonnormative approximation to $j_{1,1}$. Define the collision-free peak-normalized function

$$\mathcal{J}(x) = \begin{cases} 2J_1(x)/x, & x \neq 0, \\ 1, & x = 0. \end{cases} \quad (2)$$

For $r'_a = r/s_a$, the generic SCI-JINC coefficient is

$$\kappa_a(r'_a; a_a, b_a, c_a, (r_{\max})_a) = \mathcal{J}\left(\frac{2\pi r'_a}{a_a}\right) \exp\left[-\left(\frac{2r'_a}{b_a}\right)^{c_a}\right] \mathcal{J}\left(\frac{j_{1,1}r'_a}{(r_{\max})_a}\right). \quad (3)$$

Equation (3) has $\kappa_a(0) = 1$. a_a scales the first JINC argument, b_a scales the generalized-exponential envelope, c_a is its exponent, and $(r_{\max})_a$ places the first zero of the second JINC factor. Analytic zeros and finite signed lobes are retained. The square-cache rule, rather than a radial cutoff, determines membership.

12.2 Exact discrete point-phase and WCS operator

Let the exact AST continuous FITS pixel coordinate in the target JINC WCS be $\mathbf{u}_i = (x_i, y_i)$. Integer FITS coordinates denote pixel centers. The exact rounded cache center is

$$\mathbf{c}_i = \left(\lfloor x_i + \frac{1}{2} \rfloor, \lfloor y_i + \frac{1}{2} \rfloor\right). \quad (4)$$

Thus an exact half-pixel tie rounds toward the positive coordinate axis. The scientific owner retains this directional convention as part of the selected discrete operator. The residual phase is

$$\phi_i = \mathbf{u}_i - \mathbf{c}_i \in \left[-\frac{1}{2}, \frac{1}{2}\right)^2. \quad (5)$$

For $n_{\text{sub}} \geq 1$, each axis uses the exact half-open bins

$$B_q = \left[-\frac{1}{2} + \frac{q}{n_{\text{sub}}}, -\frac{1}{2} + \frac{q+1}{n_{\text{sub}}}\right), \quad q \in \{0, \dots, n_{\text{sub}} - 1\}, \quad (6)$$

$$q(\phi) = \lfloor n_{\text{sub}}(\phi + \frac{1}{2}) \rfloor, \quad (7)$$

$$\hat{\phi}_q = -\frac{1}{2} + \frac{q + \frac{1}{2}}{n_{\text{sub}}}. \quad (8)$$

The two axis indices form $\mathbf{q}_i$, and their midpoint representatives form $\hat{\phi}_i$. Every positive integer n_{sub} is permitted under **SCI-JINC-DEC-PHASE-CENTER-001**. The half-open upper-bin convention is observable: when n_{sub} is even, an exact $\phi = 0$ selects $q = n_{\text{sub}}/2$ and $\hat{\phi} = +1/(2n_{\text{sub}})$, with the sign interpreted in the exact target-WCS axis orientation. No special case substitutes a zero representative. The exact logical cache mapping is

$$\beta_i = q_{i1} + n_{\text{sub}}q_{i2}, \quad \mathcal{K}_{id} = (\beta_i, d_1, d_2), \quad (9)$$

equivalently $(q_{i1}, q_{i2}, d_1, d_2)$ with axis 1 varying fastest. Physical storage may use any proven value-preserving bijection of this logical key.

Base v0.1 requires an affine tangent-plane target WCS with exact pixel matrix $\mathbf{M}$ satisfying

$$\mathbf{M}^T \mathbf{M} = \Delta^2 \mathbf{I}, \quad \Delta > 0. \quad (10)$$

The exact $\mathbf{M}$ retains axis order, signs, rotation, and handedness; Equation (10) prohibits representing an anisotropic or skewed metric by a scalar pixel size. With finite destination domain $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$, define

$$h_a = \left\lceil \frac{s_a(r_{\text{max}})_a}{\Delta} \right\rceil, \quad (11)$$

$$\mathbf{d} \in \{-h_a, \dots, h_a\}^2, \quad \mathbf{p} = \mathbf{c}_i + \mathbf{d}. \quad (12)$$

For every retained square offset, the exact discrete point-phase radius is

$$\hat{r}_{id} = \left\| \mathbf{M}(\mathbf{d} - \hat{\phi}_i) \right\|_2, \quad \tilde{r}_{id} = \hat{r}_{id}/s_a, \quad (13)$$

and κ_{ip} is Equation (3) evaluated at $\tilde{r}_{id}$. The occurrence passes the center-domain gate only when $\mathbf{c}_i \in \mathcal{D}$; a square pixel is retained only when $\mathbf{p} \in \mathcal{D}$. These integer-domain predicates, rather than footprint overlap or a radial cutoff, determine geometry membership.

For an occurrence at an exact integer pixel center, both residual-phase components are zero. When n_{sub} is even, the upper-bin rule in **SCI-JINC-DEC-PHASE-CENTER-001** and Equation (10) therefore give, at the nominal central offset $\mathbf{d} = \mathbf{0}$,

$$\hat{r}_{i\mathbf{0}} = \frac{\Delta}{\sqrt{2} n_{\text{sub}}},$$

$$\kappa_{i,\mathbf{c}_i} = \kappa_a \left(\frac{\Delta}{\sqrt{2} n_{\text{sub}} s_a}; a_a, b_a, c_a, (r_{\text{max}})_a \right).$$

The nominal central cache element is thus evaluated at this exact nonzero radius and shall not be replaced by the zero-radius limit $\kappa_a(0) = 1$. For odd n_{sub} , exact zero phase has midpoint representative zero and the nominal central offset evaluates the analytic limit at zero.

12.3 Signed estimator

For the exact occurrence population of one bundle,

$$w_{ip} = \omega_i \kappa_{ip}, \quad (14)$$

$$N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i, \quad (15)$$

$$C_p = \sum_i I_{ip} \omega_i \kappa_{ip}, \quad (16)$$

$$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2, \quad (17)$$

$$m_p = \frac{N_p}{C_p}, \quad (18)$$

$$A_{pi} = \frac{I_{ip} \omega_i \kappa_{ip}}{C_p}. \quad (19)$$

All coupled accumulator terms for one contribution use the same I_{ip} and the same realized κ_{ip} identity. The normalization is signed. A finite negative C_p is admissible; exact $C_p = 0$ is not a zero-sky result.

12.4 Conditioning and local support

Define construction-state quantities

$$D_p = \sum_i I_{ip} |\omega_i \kappa_{ip}|, \quad (20)$$

$$\rho_p = \frac{|C_p|}{D_p}, \quad D_p > 0. \quad (21)$$

For exact arithmetic, $0 \leq \rho_p \leq 1$. Accepted local support requires the algebraic predicate

$$\text{AlgSupport}_p = [\text{required inputs and } N_p, C_p, Q_p \text{ are finite}] \wedge [Q_p > 0] \wedge [C_p \neq 0]. \quad (22)$$

A finite-precision numerical realization additionally requires

$$\text{NumAdeq}_p = [\mathcal{E}_p \text{ is a valid certificate under } \mathcal{P}_{\text{NA}}] \wedge [\mathcal{E}_p \models \mathcal{P}_{\text{NA}}]. \quad (23)$$

The exact profile supplies the observable, norm, domain, error propagation, near-zero rule, simultaneous or worst-case interpretation, and pass/fail inequality. No such owner-approved profile or matching certificate is admitted or supplied by this source packet, so Equation (23) is presently typed unavailable. In particular, no approximate fidelity scale promotes a near-cancellation pixel. D_p remains construction state rather than a numerical map-plane role; any future certificate is referenced by the compact generative record G_a . A passing $\mathcal{P}_{\text{NA}}/\mathcal{E}_p$ pair establishes only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over the profile's declared domain. It does not establish scientific adequacy of s_a , a_a , b_a , c_a , $(r_{\text{max}})_a$, n_{sub} , or square truncation; source-response or beam fidelity; photometric or astrometric performance; optimality; observational validation; response/covariance fidelity; readiness; or production authority. Qualification of a TolTEC parameter set is a separate scientific and observational program.

12.5 Coefficient-squared temporal accounting

The sole authorized base-v0.1 coefficient-squared temporal-accounting product is `jinc_coefficient_squared_time`, defined by

$$T_p^{(\kappa^2)} = \sum_i I_{ip} \frac{\kappa_{ip}^2}{f_{s,i}}. \quad (24)$$

It is nonnegative and has unit seconds. The squared object is κ_{ip} , not ω_i or w_{ip} .

12.6 Conditional mathematics retained for future science

The following identities do not create base-v0.1 products. For an exact available realized PTC-grid source-template response, the fixed-state response is

$$\mathbf{R}_{\text{JINC, fixed}} = \mathbf{A}_{\text{JINC}} \mathbf{H}_{\text{PTC, fixed}}. \quad (25)$$

The exact signal membership is used and the already realized PTC companion receives the JINC operator exactly once. A PTC full-procedure finite difference with JINC held fixed, a JINC re-resolved procedure response, and a whole-chain response are distinct families and must not be substituted for Equation (25) or for one another.

If and only if an exact PTC family establishes $\omega_i = \text{Var}(z_i)^{-1}$ for mutually independent admitted occurrences,

$$\text{Var}_{\text{formal}}(m_p) = \frac{Q_p}{C_p^2}, \quad (26)$$

$$W_{\text{formal}, p} = \frac{C_p^2}{Q_p}, \quad (27)$$

$$\text{Cov}(m_p, m_{p'}) = \frac{\sum_i I_{ip} I_{ip'} \omega_i \kappa_{ip} \kappa_{ip'}}{C_p C_{p'}}. \quad (28)$$

For a general available upstream covariance,

$$\mathbf{C}_{\text{JINC}} = \mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^T. \quad (29)$$

Inverse-square units alone do not establish inverse variance, independence, precision, covariance authority, or significance. A covariance state, if a future use authorizes one, is exactly one of complete numerical, partial, symbolic, summarized, lineage-resolvable, or unavailable and declares every omitted term. Missing covariance is never zero or independence. Response, variance, formal-weight, and covariance companions remain deferred until a concrete scientific use separately authorizes them.

13 Assumptions, validity classes, and explicit unavailability

Table 6: Normative assumptions and boundary states.

ID	Statement
SCI-JINC-ASM-001	The contract concerns one observation at a time and a nonpolarimetric total-intensity-equivalent transformed signal. It does not authorize a physical Stokes interpretation from a component token.
SCI-JINC-ASM-002	PTC and AST meanings, identities, validity facts, causes, uncertainty, and immutable ancestry are producer-owned facts. JINC uses them without reconstructing or redefining them.
SCI-JINC-ASM-003	An admitted ω_i is finite and strictly positive and comes from one exact PTC family/version that explicitly permits SCI-JINC. No registered family or realized payload is admitted or supplied by this source packet.
SCI-JINC-ASM-004	Equations (25)– (29) are conditional identities only. Their assumptions are not inferred from units, names, or another consumer’s permission.
SCI-JINC-ASM-005	The analytic family and parameter meanings are approved, but no TolTEC numerical realization of s_a , a_a , b_a , c_a , $(r_{\max})_a$, target-WCS metric, pixel scale, or <code>subpixel_n</code> is scientifically admitted, supplied, or recommended by this source packet.
SCI-JINC-ASM-006	Point-phase evaluation and fully populated square support are the scientific operator. Decision SCI-JINC-DEC-PHASE-CENTER-001’s upper-bin phase lattice admits every positive n_{sub} , retains the positive midpoint for exact zero phase when n_{sub} is even, and retains positive-axis half-pixel center ties. Pixel-area integration, a zero-phase special case, a circular cutoff, footprint-overlap admission, and JINC-then-crop equivalence are not alternative branches.
SCI-JINC-ASM-007	Numerical-realization adequacy requires one exact owner-approved $\mathcal{P}_{\text{NA}}$ and a matching certificate $\mathcal{E}_p$. Neither is admitted or supplied by this source packet. Finite arithmetic, analytic-function, phase/cache, and accumulation errors therefore have no authorized approximate substitute, and numerical support is typed unavailable even though exact-arithmetic algebraic support is defined. A future passing profile/certificate pair establishes only finite-precision fidelity to its exact selected discrete oracle; it does not qualify TolTEC parameters, phase resolution, square truncation, response, performance, validation, readiness, or production use.
SCI-JINC-ASM-008	Same-observation chunking may be an engineering realization of one bundle, but a chunk is not a scientific product identity. Cross-observation combination is outside base v0.1.
SCI-JINC-ASM-009	This implementation-independent authority has inspected no implementation candidate, configuration, product, test, audit, validation, reduction, or production evidence.

Every missing fact material to this authority is classified in Table 7.

Table 7: Missing-fact classification.

Class	Fact	Consequence
Normatively specified in this authority	Generic analytic identity, signed estimator, exact point-phase square operator, center-domain gate, algebraic support, fixed five-plane bundle, grouping, and ownership boundary.	These are binding frozen content in v0.1 but are not implementation or validation claims.
Not admitted or supplied by this source packet	A registered, selected, and realized JINC-permitted PTC coefficient family and payload.	No numerical bundle is formed without it; no unity or alternate-family fallback.
Not admitted or supplied by this source packet	A scientifically authorized TolTEC array-associated numerical parameter set.	No numerical bundle is formed; no inherited scale, shape, extent, or mode-dependent value is used.
Not admitted or supplied by this source packet	An exact owner-approved numerical-adequacy profile and a matching lifecycle-bound certificate.	Algebraic support remains defined, but numerical-realization adequacy, including near-cancellation support, is typed unavailable.
Implementation unassessed	Representation, interfaces, algorithms, storage, parallel realization, and implementation behavior.	No conformity or representation-fidelity claim.
Validation not performed	Algebraic execution, numerical adequacy, observational response, covariance, photometry, performance, readiness, and production operation.	No achieved result, readiness, or production claim.

Response, covariance, formal weight, physical exposure, cross-observation coaddition, standalone support or availability, diagnostics, dense per-contribution provenance, and generalized provenance products are deferred or outside base v0.1, rather than silently missing members of the fixed five-plane bundle. The compact replay record G_a is required bundle-level information state, not an additional product role or generalized provenance framework.

14 Binding scientific requirements

The key words *shall*, *shall not*, *required*, and *unavailable* are normative. These requirements define scientific conformance; they do not assert that any implementation satisfies them.

SCI-JINC-REQ-001. SCI-JINC shall define a signed spatial-kernel estimator for one observation and stable array. It shall consume exact PTC and AST parents directly, consume no SCI-MAP product, and inherit no ordinary SCI-MAP normalization, support, exposure, validity, response, covariance, product, or coaddition rule by analogy.

SCI-JINC-REQ-002. Every candidate contribution shall bind one exact occurrence and the `SCI-AST:rtc_output_grid_coordinates@1` continuous coordinate under `SCI-AST_TO_SCI-JINC v0.1/r0.2`, associated with the same processed sample realization in the exact target JINC WCS. Row order, detector order, nearest time, tolerance, shape, cardinality, or numerical coordinate equality shall not establish that association.

SCI-JINC-REQ-003. The transformed signal z_i shall retain its exact PTC quantity, unit, availability, application generation, retention state, immutable parents, nonrestored additive reference, removed-component and null-space state, response/covariance producer facts when present, and producer causes. SCI-JINC shall spatially transform that quantity without creating calibration, conditioning, astrometry, empirical noise, or source interpretation.

SCI-JINC-REQ-004. Every ω_i shall come from one exact family/version in the single PTC-owned positive analysis/gridding coefficient registry. The family shall explicitly permit SCI-JINC, and its requested, effective, observation-resolved, and realized identities, payload, QC, normalization, support, provenance, covariance

meaning, uncertainty, and prohibited meanings shall remain bound. The accepted request shall select an explicitly permitted family, or one exact owner-authorized versioned mode policy shall select it; no ambient or implementation default is authorized.

SCI-JINC-REQ-005. A numerical route shall fail closed when selection by the accepted request or an exact owner-authorized versioned mode policy is missing, the family is unregistered, JINC permission is absent, or the payload/QC/identity is absent, unavailable, incompatible, duplicate, or ambiguous. SCI-JINC shall not infer unity, another family, ordinary MAP permission, **sens**, loading, scatter, inverse variance, precision, or significance.

SCI-JINC-REQ-006. After the producer-owned coefficient decision, SCI-JINC shall admit only a finite strictly positive ω_i . Zero, negative, non-finite, or unrepresentable upstream coefficient values shall not be deposited and shall retain the exact profile-defined disposition and cause.

SCI-JINC-REQ-007. Every numerical kernel route shall bind one exact scientifically authorized array-associated parameter-set identity containing finite positive s_a , a_a , b_a , c_a , $(r_{\max})_a$, effective integer `subpixel_n` ≥ 1 with no parity restriction under the upper-bin phase-lattice rule in **SCI-JINC-DEC-PHASE-CENTER-001**, and compatibility with the exact target-WCS metric. Base v0.1 requires an affine tangent-plane WCS whose exact pixel matrix obeys Equation (10). A scalar pixel size shall not represent an anisotropic or skewed geometry. The exact frame, tangent projection, reference point, axis order and signs, rotation, handedness, angular unit, dimensions, and finite nonwrapping topology shall remain bound. Missing or incompatible authority shall make the affected route unavailable without fallback.

SCI-JINC-REQ-008. The peak-normalized analytic primitive shall be $\mathcal{J}(x) = 2J_1(x)/x$ for $x \neq 0$ with the removable limit $\mathcal{J}(0) = 1$, as in Equation (2).

SCI-JINC-REQ-009. The analytic coefficient shall follow Equation (3), including its fixed parameter ordering and $r'_a = r/s_a$. It shall have peak one, preserve analytic zeros, and preserve every finite positive and negative lobe of the complete product. The second-factor zero shall use the exact root $j_{1,1} = \min\{x > 0 : J_1(x) = 0\}$ of Equation (1); a rounded decimal shall not be its normative definition.

SCI-JINC-REQ-010. The sample center, half-pixel tie, residual-phase domain, half-open phase bins, phase-to-cache index, and midpoint representatives shall be exactly Equations (4)–(9), independently on the two exact WCS axes. Every positive integer n_{sub} is permitted. For even n_{sub} , exact zero phase shall select the positive representative $+1/(2n_{\text{sub}})$ under the bound WCS axis orientation. The selected nominal central offset shall therefore use the exact even-lattice center formulas following Equation (13), not the zero-radius limit. The positive-axis half-pixel center tie and upper-bin phase convention are the scientific-owner-approved parts of **SCI-JINC-DEC-PHASE-CENTER-001**. The selected kernel matrix shall be point evaluated and shall not be replaced by pixel-area integration.

SCI-JINC-REQ-011. Center rounding, tie handling, phase intervals, bin edges, representatives, phase-to-cache indexing, WCS metric, cache extent, destination evaluation point, and edge membership are fixed mathematical parts of the operator and shall not be implementation choices. Bessel evaluation, summation algorithm, cache storage, thread order, and parallel reduction may be engineering choices only when they realize the exact operator and carry the numerical-adequacy state required by Equations (22)–(23).

SCI-JINC-REQ-012. $(r_{\max})_a$ shall place the exact first zero of the second JINC factor and shall fix $h_a = \lceil s_a(r_{\max})_a/\Delta \rceil$. The offsets shall be exactly $\{-h_a, \dots, h_a\}^2$, and each coefficient shall use the destination pixel-center and representative-phase radius of Equation (13). Every resolved square pixel, including a corner with radius greater than $(r_{\max})_a$, shall be evaluated; no circular cutoff or radial membership predicate shall be applied.

SCI-JINC-REQ-013. Before any footprint evaluation, the exact rounded cache center of Equation (4) shall lie in $\mathcal{D} = \{1, \dots, N_x\} \times \{1, \dots, N_y\}$. An outside center shall set $I_{ip} = 0$ for every pixel, even when its square would overlap the map; footprint-overlap admission is prohibited.

SCI-JINC-REQ-014. For an admitted in-map center, the exact square pixel $\mathbf{p} = \mathbf{c}_i + \mathbf{d}$ shall be retained if and only if $\mathbf{p} \in \mathcal{D}$ and every other local gate passes; square pixels outside the finite map shall be absent. The route shall not wrap, reflect, complete the footprint, substitute an interior normalizer, renormalize the edge, or require JINC-then-crop equivalence.

SCI-JINC-REQ-015. Every evaluated κ_{ip} shall be finite. A finite negative coefficient is normal and shall retain its sign; an analytic zero is a contract-defined zero contribution distinct from outside support; a non-finite coefficient shall fail the affected required bundle before deposition rather than be clipped, skipped, or zero-filled.

SCI-JINC-REQ-016. All coupled accumulator terms for one contribution shall use the same exact occurrence, AST association, sample-admission decision, rounded-center decision, sample-pixel membership, phase, and κ_{ip} identity.

SCI-JINC-REQ-017. The required accumulators shall be distinct and shall obey Equations (15)–(17). Negative lobes shall keep their sign in N_p and C_p and enter Q_p quadratically.

SCI-JINC-REQ-018. The published signal estimate shall be $m_p = N_p/C_p$ and the fixed-state normalized operator shall be Equation (19). No Q_p denominator or other local measurement model shall be substituted.

SCI-JINC-REQ-019. Algebraic local support shall be exactly Equation (22): finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, and $C_p \neq 0$. This state shall remain distinct from certified finite-precision numerical adequacy. A rejected pixel shall carry local invalid support in the `jinc_map` role and shall not receive a finite substitute.

SCI-JINC-REQ-020. A finite negative C_p shall be admissible. Exact $C_p = 0$ shall be locally invalid and unavailable, never interpreted as zero sky, infinity, or a valid non-finite map value.

SCI-JINC-REQ-021. SCI-JINC shall retain the dimensionless conditioning statistic ρ_p of Equation (21). It is an indicator and shall not itself promote numerical support. A finite-precision result shall be numerically supported only when an exact owner-approved $\mathcal{P}_{NA}$ and matching $\mathcal{E}_p$ satisfy Equation (23); otherwise numerical adequacy, including near-cancellation support, shall be typed unavailable.

SCI-JINC-REQ-022. No ambient universal ρ_p cutoff, unit-bearing C_p or Q_p floor, contributor-count/machine-epsilon formula, prescribed summation algorithm, fixed reduction order, bitwise reproducibility, approximate fidelity phrase, or implementation default shall be treated as scientific authority. A future exact numerical-adequacy profile may define only its explicitly bounded domain-specific predicate.

SCI-JINC-REQ-023. Every claimed numerical-support result shall bind one exact owner-approved numerical-adequacy profile and matching certificate. Together they shall identify the scientific observable; absolute or relative form; reference magnitude or norm; parameter, source, and pixel domain; numerator and denominator error treatment; near-zero behavior; simultaneous or worst-case interpretation; exact pass/fail inequality; uncertainty; lifecycle; and provenance. No such profile or certificate is admitted or supplied by this source packet, so no numerical-support result is presently authorized. A passing pair shall establish only finite-precision fidelity to the exact selected discrete SCI-JINC oracle over its declared domain. It shall not establish scientific adequacy of the kernel parameters, n_{sub} , or square truncation; source-response or beam fidelity; photometric or astrometric performance; optimality; observational validation; response/covariance fidelity; readiness; or production authority.

SCI-JINC-REQ-024. The unit relations in Table 5 and the two allowed ω_i unit classes shall be preserved. Common positive scaling of all ω_i shall leave m_p , ρ_p , and A_{pi} invariant; signal-unit rescaling shall transform the estimator consistently.

SCI-JINC-REQ-025. Equations (26)–(29) may receive a formal statistical interpretation only when the exact PTC family and covariance assumptions establish it. Inverse-square units or historical labels alone shall not establish variance, precision, covariance, or significance. Any future covariance state shall be exactly typed as complete numerical, partial, symbolic, summarized, lineage-resolvable, or unavailable, shall declare omitted terms, and shall never interpret missing covariance as zero or independence.

SCI-JINC-REQ-026. The required coefficient-squared temporal-accounting product `jinc_coefficient_squared_time` shall be exactly Equation (24). It shall square κ_{ip} , not ω_i or w_{ip} , and shall have unit seconds.

SCI-JINC-REQ-027. `jinc_coefficient_squared_time` shall be described only as method-specific coefficient-squared temporal accounting. It shall not be interpreted as physical acquired exposure, valid-original exposure, complete temporal support, hits, normalized influence, white-noise-equivalent time, precision, validity, confidence, or significance.

SCI-JINC-REQ-028. Every produced base-v0.1 bundle shall contain exactly five required numerical map-plane roles: `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, derived `jinc_map` with local support/validity, and `jinc_coefficient_squared_time`. The compact bundle-level generative record required by SCI-JINC-REQ-044 shall be information state at the existing plan/bundle provenance granularity and shall not be a sixth product role.

SCI-JINC-REQ-029. Failure to form or publish any required whole-product role shall suppress the affected bundle atomically. Pixel-local zero, insufficient, cancelled, or otherwise invalid support shall remain ordinary bundle content and shall not suppress an otherwise complete role or create a role-availability object.

SCI-JINC-REQ-030. One observation may produce zero through three independent bundles, with at most one for each stable array admitted and requested under the exact JINC realization. A missing, unavailable, or unrequested array shall create no placeholder or empty-array product and shall not invalidate another array's bundle.

SCI-JINC-REQ-031. Each bundle shall be independently identified by observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Its compact generative record G_a shall bind those identities without making a worker, filename, or dense contribution archive part of scientific identity. Contributions with different array or destination identities shall never merge.

SCI-JINC-REQ-032. The complete destination identity shall be resolved before allocation, numerical mutation, output side effects, or parallel execution. Ambiguous, duplicate, or aliased destination ownership shall fail the affected bundle without selecting a winner or publishing a partial result.

SCI-JINC-REQ-033. Same-observation samples or chunks may accumulate incrementally only when observation, stable array, JINC plan and realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation are identical and are replayable from the same G_a . Chunk, worker, process, container, filename, and vector position shall not create product identity.

SCI-JINC-REQ-034. Base v0.1 shall authorize no cross-observation combination. Any future JINC coadd shall require a separately authorized boundary over complete observation bundles and shall not infer accumulator addition, normalized-map combination, or ordinary SCI-MAP coaddition.

SCI-JINC-REQ-035. JINC upstream-occurrence admission, whose immutable Registry identity is `SCI-JINC:jinc_map_contribution@1`, shall remain distinct from PTC signal availability, PTC retention, coefficient availability/QC, AST coordinate validity and association, finite signal, finite positive ω_i , parameter availability, rounded-center admission, pixel support, finite κ_{ip} , cancellation, numerical adequacy, arithmetic contribution, and final bundle validity. The immutable Registry identity shall not routinely be described as a “map contribution” decision; its canonical human-readable meaning is only that one exact upstream occurrence may be considered by JINC.

SCI-JINC-REQ-036. Only a fact named by an exact restriction shall control JINC upstream-occurrence admission. Producer causes and transitive influence shall be preserved, but a generic flag or cause shall have no universal veto. Outside support, an analytic zero, or an outside rounded center shall be ordinary no-contribution; a finite negative lobe shall not create a cause. Profile pass shall not imply rounded-center admission, pixel membership, cancellation support, numerical contribution, or bundle publication.

SCI-JINC-REQ-037. ALIGN/AST, RTC, CAL, PTC, and VAL shall retain the meanings they own. SCI-JINC shall own only its admission use, local geometry, point phase, square support, signed coefficient, normalization, conditioning, fixed accumulators, local support, grouping, destination, and atomic bundle publication. The exact PTC r0.3 and AST r0.2 boundaries, immutable admission profile, and JINC-specific SCI-VAL successor snapshots shall control compatibility under the exact-source lock of SCI-JINC-REQ-043; SCI-JINC shall not reconstruct their producer facts.

SCI-JINC-REQ-038. Base v0.1 shall publish no response, variance, formal-weight, covariance, empirical-noise, or significance product. The conditional mathematics in Section 12.6 shall remain preserved for a future concrete scientific use without being promoted to a present role. Fixed-state JINC response, PTC full-procedure response with JINC fixed, JINC re-resolved procedure response, and whole-chain response shall remain distinct; an already realized PTC-grid response shall receive the JINC operator exactly once and shall use exact signal membership.

SCI-JINC-REQ-039. Base v0.1 shall publish no physical-exposure product. Any future exposure role shall require separate authority for exact original- occurrence lineage and semantics and shall not reinterpret $T_p^{(\kappa^2)}$.

SCI-JINC-REQ-040. Base v0.1 shall create no standalone support or availability role, generic optional or conditionally required role mechanism, detailed operational-cause archive, diagnostic role, generalized provenance product, or per-pixel or per-contribution provenance system. This prohibition shall not remove the compact bundle-level generative record required by **SCI-JINC-REQ-044**.

SCI-JINC-REQ-041. No TolTEC numerical parameter realization, recommendation, optimum, hidden default, or fallback is authorized by v0.1. In particular, an inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values shall not be treated as scientific authority.

SCI-JINC-REQ-042. A future conformance assessment shall exercise every prediction in Section 15, bind any exact owner-approved numerical- adequacy profile and certificate rather than an approximate acceptance phrase, and separate algebraic correctness from implementation conformity, representation/response fidelity, numerical and observational validation, achieved performance, readiness, and production authorization. A passing profile/certificate pair shall be reported only as exact-discrete-oracle finite-precision fidelity under **SCI-JINC-REQ-023**. This frozen implementation-independent authority establishes none of those later claims.

SCI-JINC-REQ-043. Every claimed SCI-JINC contract realization shall bind byte identically to SCI-PTC_TO_SCI-JINC v0.1/r0.3 (SHA-256 5769d413460e931745e0d401ea432b12d1077c15466247c49caa71b997d4ab1e), SCI-AST_TO_SCI-JINC v0.1/r0.2 (SHA-256 eeffa7059b59c89793fa1d523fb3bb48235f1ab55f7d55060af1600cbfd470a5), the SCI-JINC:jinc_map_contribution@1 profile (SHA-256 2db95da7e5d1b980df79993907d45ac0ababc3aa05c189bfb62dcf04ff2c2e8a), and the SCI-VAL binding record (SHA-256 ee8f20db5febdb51e39f7157449d6c2d03a0d17058605dd5531e9ab5ca439e30). The bound SCI-VAL source-register successor shall be SCI-VAL_SOURCE_BINDING_REGISTER version v0.1/r0.3-map-r0.7.1-jinc-stage-a-q002-2026-08-28 with SHA-256 0e7ca29ee2e9cd02fb1b76cf87cc64fce6164407a7801f9b9a105ca646317e88, and the profile-registry successor shall be SCI-VAL_PROFILE_REGISTRY version v0.1/r0.3-map-r0.7.1-jinc-stage-a-q002-2026-08-28 with SHA-256

4b9a1ebecfc847c83b59da772afd9b031ab1830e8febbb12d1a47f70ce5a1110. The SCI-VAL successor identities and hashes shall be process bindings of the exact source/profile bytes only; they shall create no SCI-JINC scientific content, PTC coefficient family, TolTEC parameter set, or numerical-adequacy profile. Base v0.1 uses an exact-source-lock lifecycle: the complete exact files named above, including the JINC-stage-A-Q002 source/profile successor snapshots, are the bound objects. No ambient “current Registry” or nearby row/profile is consulted, and an unrelated Registry state does not replace these snapshot bytes. Any changed bound byte, identity, version, compatibility, or supersession fact shall make the affected route unavailable and requires a versioned SCI-JINC successor before use.

SCI-JINC-REQ-044. Every complete bundle shall contain one compact generative record G_a sufficient to replay its exact input population and immutable PTC/AST joins; admission evaluation; coefficient family, selection, generation, and payload identity; parameter-set identity; target WCS and metric; center, phase, bin, representative, cache, square, and edge rules; cancellation and numerical- adequacy state; destination identity; requested, effective, resolved, and realized lifecycle; decision **SCI-JINC-DEC-PHASE-CENTER-001**; the exact-source-lock identities; every required-role state; failure; and completion. The record shall use immutable references and bounded plan/bundle information state, not dense per-contribution provenance or a new product role.

15 Falsifiable predictions and edge cases

These predictions are normative tests for a future authorized conformance program. Listing them is not evidence that they have been executed or passed.

Table 8: Stable SCI-JINC predictions.

Prediction	Falsifiable expected result
SCI-JINC-PRED-001	A constant finite input $z_i = z_0$ returns $m_p = z_0$ on every accepted local support, including a cropped edge pixel.
SCI-JINC-PRED-002	One admitted contributor with nonzero finite κ_{ip} returns $m_p = z_i$, whether the coefficient is positive or negative.
SCI-JINC-PRED-003	A hand-computed two-contributor case with unequal z_i , ω_i , and signed κ_{ip} matches the separate N_p , C_p , Q_p , m_p , and $T_p^{(\kappa^2)}$ equations.
SCI-JINC-PRED-004	Multiplying every ω_i by the same positive factor scales N_p , C_p , and Q_p together and leaves m_p , ρ_p , and A_{pi} unchanged.
SCI-JINC-PRED-005	Under a true inverse-variance family, rescaling $z_i \mapsto \lambda z_i$ and $\omega_i \mapsto \omega_i/\lambda^2$ produces $m_p \mapsto \lambda m_p$ and conditional formal weight $W_{\text{formal},p} \mapsto W_{\text{formal},p}/\lambda^2$.
SCI-JINC-PRED-006	The analytic evaluator returns $\kappa_a(0) = 1$ through the removable limit, not a non-finite value or an implementation-dependent substitute.
SCI-JINC-PRED-007	At $r'_a = (r_{\max})_a$, the exact identity $J_1(j_{1,1}) = 0$ makes the second JINC factor zero. The resulting analytic $\kappa_{ip} = 0$ changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$ while remaining distinguishable from outside square support; a rounded decimal is not the root oracle.
SCI-JINC-PRED-008	A finite negative lobe retains its sign in N_p and C_p , enters Q_p and $T_p^{(\kappa^2)}$ quadratically, and is not clipped, absolutized, omitted, or flagged solely for being negative.
SCI-JINC-PRED-009	Exact cancellation $C_p = 0$ yields local invalid support and no finite substitute, even when N_p and Q_p are finite.
SCI-JINC-PRED-010	A finite negative C_p with otherwise adequate support yields the signed normalized estimate and is not rejected merely for the denominator sign.
SCI-JINC-PRED-011	A near-cancellation construction is numerically eligible only when a certificate $\mathcal{E}_p$ passes the exact inequality and complete domain rules of the bound owner-approved $\mathcal{P}_{\text{NA}}$; ρ_p alone neither promotes nor rejects it. A pass establishes only finite-precision fidelity to the exact selected discrete oracle on that domain and establishes none of the parameter, truncation, response, beam, photometric, astrometric, optimality, validation, fidelity, readiness, or production claims excluded by SCI-JINC-REQ-023.
SCI-JINC-PRED-012	With no exact owner-approved $\mathcal{P}_{\text{NA}}$ and matching $\mathcal{E}_p$, numerical adequacy is typed unavailable, even when algebraic support holds and finite arithmetic happens to return a nonzero C_p .
SCI-JINC-PRED-013	A non-finite required signal, geometry, coefficient evaluation, accumulator, or sample frequency cannot be masked by multiplication by zero or silently skipped; the exact local or whole-bundle failure rule applies.
SCI-JINC-PRED-014	Square-cache corners whose radius exceeds $(r_{\max})_a$ are evaluated by Equation (3) and may contribute.
SCI-JINC-PRED-015	Two points on opposite sides of a hypothetical circular $r_{\max}$ boundary but inside the same resolved square are both governed by square membership; no radial cutoff appears.
SCI-JINC-PRED-016	Effective <code>subpixel_n=1</code> selects bin $q = 0$ and midpoint representative $\hat{\phi} = 0$ on each axis for every residual phase; it selects one point-phase matrix and does not become a pixel-area average.

Prediction	Falsifiable expected result
SCI-JINC-PRED-017	Coordinates immediately around every half-pixel tie and phase-bin edge select the exact centers, half-open bins, indices, and midpoint representatives of Equations (4)– (8). For even n_{sub} , exact $\phi = 0$ selects the positive representative $+1/(2n_{\text{sub}})$ in the bound WCS axis orientation. With zero phase on both axes and $\mathbf{d} = \mathbf{0}$, the exact radius is $\Delta/(\sqrt{2}n_{\text{sub}})$ and the coefficient is the exact even-lattice center formula following Equation (13), not an imposed $\kappa_a(0) = 1$. An exact positive half-pixel coordinate tie selects the larger center and residual phase $-1/2$ under SCI-JINC-DEC-PHASE-CENTER-001. No parity restriction, zero-phase special case, or alternate center-tie rule is permitted.
SCI-JINC-PRED-018	The exact rounded center immediately outside each map edge or corner changes none of the four accumulators for any pixel, even when its square footprint overlaps the map.
SCI-JINC-PRED-019	A rounded center immediately inside an edge uses only retained square pixels with no wrap, reflection, completion, renormalization, or edge correction.
SCI-JINC-PRED-020	A supported constant field at an in-map cropped edge still returns the constant through the actual retained N_p/C_p terms.
SCI-JINC-PRED-021	$T_p^{(\kappa^2)}$ is unchanged by changing ω_i alone, receives zero from an analytic zero, and receives a positive term from either sign of nonzero κ_{ip} .
SCI-JINC-PRED-022	Missing coefficient selection by the accepted request or one exact owner-authorized versioned mode policy, an unregistered family, missing JINC permission, or an unavailable/mismatched payload prevents the affected numerical bundle without an ambient, unity, or alternate-family fallback.
SCI-JINC-PRED-023	Absence of an authorized TolTEC parameter set prevents the numerical bundle; no inherited 45 m denominator, shape values, mode-dependent r_{max} , or other implementation value is substituted.
SCI-JINC-PRED-024	Permuting arrays, networks, detector rows, map groups, or container positions does not change array-to-parameter or array-to-bundle identity; a missing array creates no placeholder and does not invalidate another array.
SCI-JINC-PRED-025	Ambiguous, duplicate, or aliased destination ownership fails the affected bundle before mutation and produces no partial winner.
SCI-JINC-PRED-026	Valid decompositions of the same observation into chunks replay the same scientific population, immutable joins, memberships, and equations from G_a . No numerical agreement claim is available without an exact $\mathcal{P}_{\text{NA}}$ and matching certificates; bitwise equality is not a scientific requirement.
SCI-JINC-PRED-027	An attempted cross-observation accumulator or map combination is rejected as outside base v0.1 rather than being assigned an inferred coadd rule.
SCI-JINC-PRED-028	A missing, duplicate, or ambiguous exact signal-coordinate association under the exact PTC r0.3 and AST r0.2 boundaries prevents the affected contribution and is not reclassified as ordinary outside support or repaired by another source identity.
SCI-JINC-PRED-029	Failure to form any one of the five required whole-product roles suppresses the bundle, whereas a locally invalid pixel does not suppress an otherwise complete role.
SCI-JINC-PRED-030	The base bundle contains exactly five numerical map-plane roles and no response, covariance, formal-weight, physical-exposure, standalone availability/support, diagnostic, generalized-provenance, or placeholder product. Its compact G_a information state is present but is not counted or published as a sixth product role.
SCI-JINC-PRED-031	With dimensionless ω_i , Q_p/C_p^2 remains dimensionless and is not reported as signal variance or used to create a formal-weight product.

Prediction	Falsifiable expected result
SCI-JINC-PRED-032	The JINC route consumes no SCI-MAP product and no ordinary MAP admission, validity, exposure, support, response, covariance, or coaddition decision is substituted for a JINC-owned decision. A pass from JINC upstream-occurrence admission (Registry identity <code>SCI-JINC:jinc_map_contribution@1</code>) means only that the occurrence may be considered.
SCI-JINC-PRED-033	A changed byte, identity, version, compatibility, or supersession fact for the exact PTC r0.3 boundary, AST r0.2 boundary, admission profile, SCI-VAL binding, source-register successor, or profile- registry successor makes the affected route unavailable; no nearby source or name match is accepted. No ambient “current Registry” state is accepted; any changed bound object requires a versioned SCI-JINC successor. Matching SCI-VAL process bindings creates no new scientific content, coefficient family, parameter set, or adequacy profile.
SCI-JINC-PRED-034	Passing JINC upstream-occurrence admission (Registry identity <code>SCI-JINC:jinc_map_contribution@1</code>) leaves rounded-center admission, sample-pixel membership, cancellation support, numerical adequacy, numerical contribution, and bundle publication to their separate exact gates.
SCI-JINC-PRED-035	A target WCS whose pixel matrix is anisotropic or skewed fails Equation (10) and cannot enter base v0.1 through a scalar pixel size. A compatible matrix preserves its exact axis ordering, signs, rotation, and handedness in phase and cache interpretation.
SCI-JINC-PRED-036	Replaying the immutable references and bounded state in G_a reproduces the exact input population, PTC/AST joins, admission, coefficient and parameter identities, target metric, center/phase/bin/cache decisions, square/edge membership, destination, lifecycle, required-role states, failure, and completion without a dense per-contribution archive.

15.1 Prediction-to-requirement trace

Predictions	Governing requirements
001–003	016–020, 026
004–005	024–025
006–008	008–009, 015, 017, 026
009–012	019–023
013	003, 006–007, 015, 019, 026, 029
014–017	007, 009–012
018–020	013–014, 016–020
021	026–027
022–023	004–007, 041
024–027	030–034, 044
028	002, 035–037, 043
029–030	028–029, 038–040, 044
031	024–025, 038
032	001, 035–038
033	002, 035, 037, 043
034	013, 019–023, 035–036
035	007, 010–014
036	031, 033, 040, 044

16 Blank conformance-result template

Field	Future value
Source-byte equality	Not assessed for a candidate
Candidate/lifecycle identity	Not assessed
Coefficient family and parameter set	Not assessed
Observation, array, WCS, destination	Not assessed
Exact discrete replay	Not assessed
Requirement results (44)	Not assessed
Prediction results (36)	Not assessed
Numerical profile and certificate	Typed unavailable in this source packet
Five-role publication	Not assessed
Forbidden companion-role absence	Not assessed
Representation/response/covariance fidelity	Not assessed
Numerical/observational validation	Not performed
Achieved performance	No claim
Readiness/production authorization	No claim